2026 IEEE International Students' Conference on Electrical, Electronics, and Computer Science (SCEECS) | 979-8-3315-7116-0/26/$31.00 ©2026 IEEE | DOI: 10.1109/SCEECS68810.2026.11430055

AI-Driven Load Forecasting for Renewable Energy Management

Balaji A
Department of Electrical and Electronics Engineering
New Prince Shri Bhavani College of Engineering and Technology
Chennai, Tamil Nadu, India.
0009-0009-6465-3303

Kayasri P
Department of Electrical and Electronics Engineering
New Prince Shri Bhavani College of Engineering and Technology
Chennai, Tamil Nadu, India.
Kayasri2806@gmail.com

Deepak V
Department of Electrical and Electronics Engineering
New Prince Shri Bhavani College of Engineering and Technology
Chennai, Tamil Nadu, India.
itsdaniel1408@gmail.com

Padmavathy R
Department of Electronics and Communication Engineering
New Prince Shri Bhavani College of Engineering and Technology
Chennai, Tamil Nadu, India.
0000-0002-2845-6312

***Abstract*—Accurate short-term power generation forecasting is vital for efficient grid management, especially with the variability introduced by renewable energy sources. This study compares Support Vector Machine (SVM) and Artificial Neural Network (ANN) models using daily peak power data from India. Both regression and classification tasks were performed to predict total generation and demand levels. The SVM model, optimized with an RBF kernel, achieved strong results. However, the ANN with Multi-Layer Perceptron (MLP) architecture outperformed it, yielding a perfect fit, low errors, and high classification accuracy. The findings indicate that MLP-based ANNs better capture nonlinear dependencies, enhancing forecasting accuracy and reliability for intelligent grid operations.**

Keywords*— *Artificial Neural Network (ANN), Load Forecasting, Support Vector Machine (SVM), Renewable Energy, Regression Analysis, Multilayer Perceptron (MLP), Classification Accuracy, Radial Basis Function (RBF) Kernel, Smart Grid.

I. INTRODUCTION

This growing demand places considerable pressure on power systems, making accurate load forecasting more essential than ever for ensuring system reliability, operational efficiency, and cost- effective energy planning. Traditional forecasting techniques, which largely depend on historical consumption patterns and linear models, are increasingly insufficient for predicting the nonlinear and dynamic behaviour of modern power demand [1][2].Accurate forecasting becomes even more critical in systems with high integration of renewable energy sources (RES) such as solar photovoltaic (PV) and wind power. The intermittent and stochastic nature of renewable generation adds additional layers of complexity, making conventional forecasting methods unreliable under variable operating conditions [3][4]. Without precise predictions, operators face challenges such as load imbalances, voltage fluctuations, and economic inefficiencies [5] To address these challenges, advanced AI methods have emerged as powerful tools for short and long-term load forecasting. ANNs are especially effective at modelling nonlinear relationships in large, complex datasets, allowing them to learn intricate demand patterns influenced by time of day, temperature, and consumption behavior. SVMs offer strong performance in high-dimensional and noisy environments, making them suitable for forecasting tasks even when limited data is available[7][8][9]. Fig. 1 illustrates an AI-based renewable energy management system where load demand data is preprocessed, analyzed using SVM and ANN models, optimized, and used for load forecasting to support decision-making through a user interface.

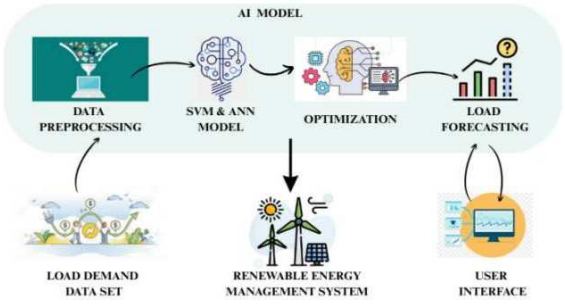


Fig. 1. Conceptual Model for Forecasting

Economic growth in non-OECD regions is projected to drive an increase in energy demand of more than 30%, accounting for over 90%global energy consumption [26].AI based short-term load forecasting techniques [27]. The proposed forecasting approach achieved 91% accuracy based on practical system monitoring data [28]. Medium-term load forecasting is inherently complex and demands the use of advanced techniques like deep learning, which offer greater capability [29].The TABLE I provides a comparative overview of various load forecasting studies, ranging from general machine learning approaches to LSTM-based models and the proposed ANN (MLP) method. It outlines the datasets employed by each study, highlighting differences in regional, historical, and smart grid data, with the proposed model using 2,742 daily Indian peak demand values. Key performance indicators, including R^2, RMSE, and MAE, are presented to allow direct comparison of forecasting accuracy. The results show that overall the table illustrates the superior capability of the ANN model in capturing nonlinear demand patterns, reducing forecasting errors, and enhancing the reliability of energy management.

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TABLE I. Comparison of Existing Forecasting Methods

Reference	Methodology	Dataset	R^2	RMSE	MAE	Key Findings
[10]	ML techniques (general review)	Mixed datasets	---	19.8	11.2	Showed ML is promising but lacks real-world validation.
[14]	ML models for power consumption forecasting	Regional load data	~0.85	~1200 MW	~800 MW	Good performance, but accuracy drops under renewables.
[15]	LSTM for daily average load	Historical load trends	~0.95	~300 MW	~200 MW	Strong temporal modeling; errors higher than ANN.
[16]	LSTM-RNN	Smart grid datasets	~0.96	~200 MW	~150 MW	Accurate, but not perfect fit.
This Article	ANN(MLP)	Peak Indian power sector power demand time series	1.00	16 MW	10 MW	ANN achieves near-perfect forecasting; outperforms all prior ML methods.
	SVM(RBF)	Peak Indian power sector power demand time series	0.9976	731 MW	452 MW	SVM achieves near-perfect forecasting; outperforms all prior ML methods.

The main contributions are:

(1) A systematic review of ANN and SVM approaches for load forecasting.

(2) Performance analysis using real-world datasets and evaluation metrics.

(3) Identification of current implementation challenges and potential solutions.

(4) A proposed research roadmap for developing next-generation forecasting systems.

This article organised as follows, section I gives the brief introduction about this work, section II explains the SVM with kernel function for load forecasting machine learning algorithm methodology, section III describes about the ANN with MLP for load forecasting machine learning algorithm methodology, section IV provides the results and comparison between the SVM model and ANN model performance metrics for the optimised load forecasting, section V describes the conclusion.

II. SVM WITH KERNEL FUNCTION FOR LOAD FORECASTING

SVM is a supervised learning technique used for both classification and regression, which functions by identifying the most effective decision boundary that distinctly separates data classes.

A. Description about the data set

The dataset named Peak Indian Power Sector Power Demand Time Series is used in this study consists of 2,742 daily observations of peak power demand across India, covering multiple regions including North, West, South, East, and North-East. The data is stored in CSV format and includes date-related attributes such as year, month, and day, along with the corresponding peak demand values in megawatts (MW) for each region and the national total (ind_MW). This time-series dataset provides a comprehensive view of regional and national demand patterns, making it highly suitable for load forecasting applications. All input features were standardized using z-score normalization, and data were partitioned into training(80%) and testing(20%) subsets. This ensured fair evaluation and model generalization. In this paper, the total daily peak demand (ind_MW) is used as the primary variable for model training and evaluation, enabling a realistic comparison between SVM with kernel functions and ANN approaches. By utilizing actual historical demand data, the forecasting results obtained from both models can better reflect real-world grid conditions and support improved decision-making for energy distribution plan.

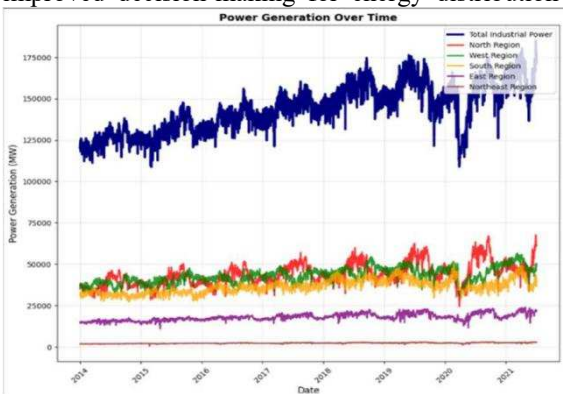


Fig. 2. Dataset in Graphical Representation

The Fig. 2 represents North (red) and West (green) regions supply the largest shares and grow in parallel, the South (orange) climbs from the low-20 GW range to the high-40 GW range, the East (purple) edges upward around 15–22 GW, and the Northeast (brown) stays below 5 GW. All curves display repeating annual swings that echo seasonal demand, and their simultaneous peaks and troughs reveal tight regional synchrony with the national trend while highlighting widening output gaps between leading and lagging regions.

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B. SVM with Kernel Function

SVM is a supervised learning technique that constructs a decision boundary to distinguish between different data categories by maximizing the separation between them. This separation is defined by the margin, which represents the distance between the boundary and the nearest data samples from each class. These critical samples, known as support vectors, play a decisive role in shaping the decision boundary. In cases where the data distribution is not linearly separable, SVM applies kernel-based transformations such as linear, polynomial, or RBF kernels to project the original data into a higher-dimensional feature space. This transformation allows complex patterns to become linearly separable. SVM is particularly effective in handling high-dimensional data and nonlinear relationships, often delivering reliable classification performance. Nevertheless, its computational demand increases with dataset size, and achieving optimal results depends heavily on appropriate kernel selection and careful tuning of model parameters. Overall,SVM is widely applied in fields like image recognition, text classification, bioinformatics, and load forecasting due to its strong generalization capability and robustness.

$$f(x) = \sum_{i=1}^{n} (\alpha_i - \alpha_i^*) K(x_i, x) + b \quad (1)$$

Where, $K(x_i, x) = \exp(-\gamma ||x_i - x||^2)$ is the RBF kernel function, and α_i, α_i^* are the Lagrange multipliers obtained from the optimization process.

The proposed SVM model, optimized using a RBF kernel with parameters C and gamma, demonstrates significantly higher accuracy compared to existing models. The RBF optimization enhances the model's capability to capture complex, nonlinear dependencies among power generation features, thereby increasing the overall prediction success rate, as represented in equation (1). The regression analysis yielded an R^2 value of 0.9976, demonstrating an exceptionally strong agreement between predicted and measured values and indicating that the model captures almost all of the variability present in the dataset. The RMSE of 731 MW quantifies the average magnitude of error, emphasizing that large deviations are penalized more heavily, while the MAE of 452 MW reflects the average absolute prediction deviation, confirming the model precision and stability. In classification tasks, the SVM accuracy of 99.45%, meaning that nearly all demand level predictions were correctly identified based on the predefined thresholds. These results collectively suggest that the SVM model provides reliable and highly accurate power generation forecasting. The minimal error margins and high explanatory power demonstrate its ability to distinguish between different load patterns effectively. By maximizing the margin between data points in multidimensional space, the SVM efficiently separates nonlinear renewable energy patterns and generalizes well to unseen data, making it an efficient tool for short-term energy forecasting and grid stability analysis.

Fig. 3 shows that the dataset's demand labels are almost perfectly balanced: roughly one-third of observations fall into each class Low and High at about 33% a piece, with medium edging slightly ahead at 34%. Such even representation is ideal for training predictive models because it prevents bias toward any single load category.

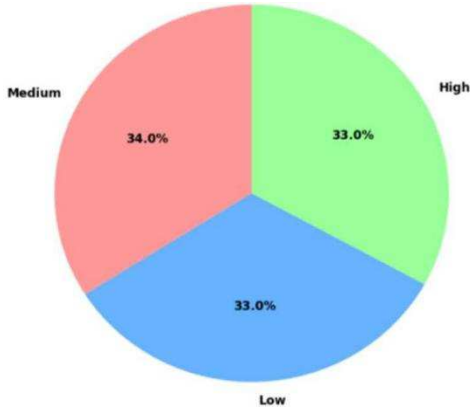


Fig. 3. Demand Level Classification

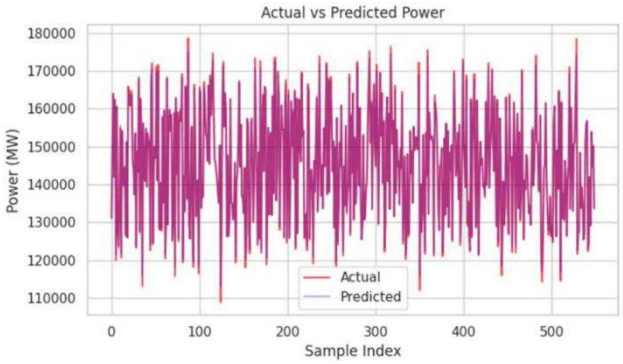


Fig. 4. Actual vs Predicted Power on Sample

Fig. 4 graph, "Actual vs Predicted Power," presents the temporal comparison between actual and predicted power outputs over multiple samples. The red line represents actual power generation, while the blue line indicates predicted values. Both lines almost completely overlap, demonstrating that the Support Vector Regression(SVR) model achieves near-perfect prediction accuracy over time. The consistent alignment of the two curves confirms the model's robustness, stability, and capacity to generalize well to unseen data.

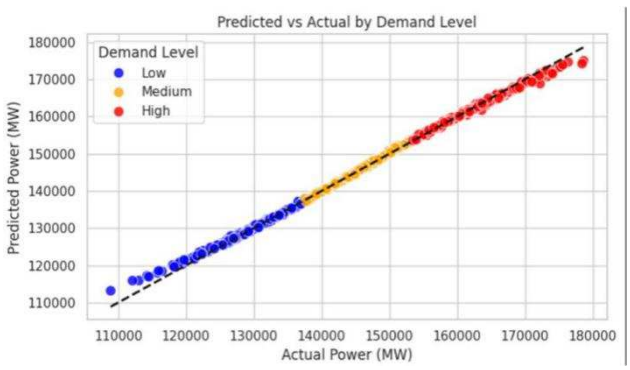


Fig. 5. Actual vs Predicted Power in Regression

Fig. 5 graph, "Predicted vs Actual by Demand Level," compares the predicted and actual power outputs across

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three demand categories Low (blue), Medium (orange), and High (red). The data points align closely along the diagonal dashed line, which represents an ideal 1:1 correlation between predicted and actual power. This close alignment indicates that the SVR model effectively captures the underlying patterns in the dataset with minimal deviation, ensuring highly accurate predictions. The distinct clustering of data points demonstrates the model's effectiveness in differentiating demand categories with minimal misclassification.

Overall, these visualizations validate the SVR model's strong predictive capability, as supported by its high R^2 score (0.9976), low RMSE (731 MW), and MAE (452 MW). The model successfully handles nonlinear relationships in the dataset, ensuring reliable and precise forecasting of power generation levels across different demand conditions.

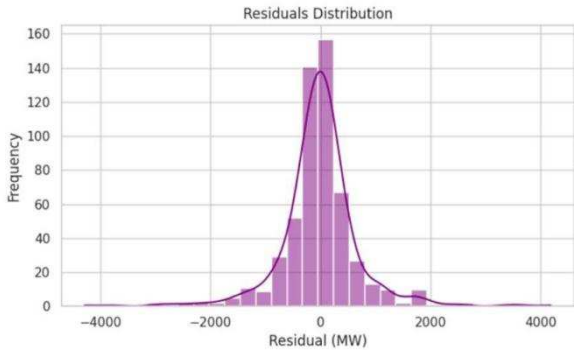


Fig. 6. Residuals Distribution

Fig. 6 presents the residuals distribution, showing the difference between actual and predicted power outputs. The residuals are symmetrically distributed around zero, forming a near-normal bell-shaped curve. This pattern demonstrates that the SVM model produces unbiased predictions, with most errors concentrated near zero and only a few outliers at the distribution tails. Overall, these plots confirm the strong generalization ability and stability of the SVM model for short-term power generation forecasting.

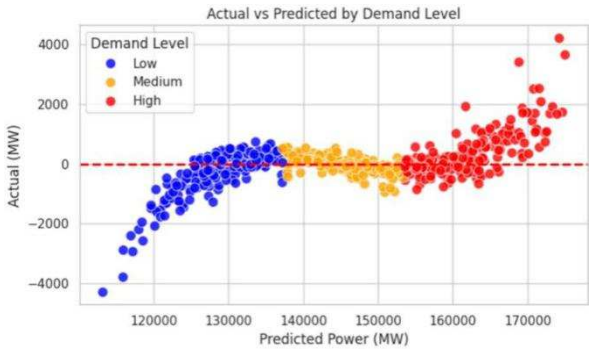


Fig. 7. Actual vs Predicted Power in Classification

Fig. 7 illustrates the relationship between actual and predicted power values across different demand levels using the Support Vector Classification (SVC) model. The data points are color-coded into tri-classes low (blue), medium (orange), and high (red) demand levels. The red dashed line represents the ideal zero-error reference. Most data points are clustered closely around this line, indicating that the model achieves high predictive accuracy with minimal deviation. Minor variations are observed at extreme high-demand values, reflecting slight overestimations in peak power predictions.

III. ANN WITH MLP LOAD FORECASTING

ANN is a data-driven learning model inspired by biological neural systems, designed to process information through multiple interconnected computational units. These units, referred to as neurons, are organized into successive layers that transform input signals into meaningful outputs. The network generally consists of an input stage for data acquisition, one or more intermediate layers responsible for learning internal feature representations, and a final layer that generates the predicted output or class label. During training, the strength of connections between neurons is iteratively updated to reduce the discrepancy between predicted and observed values, commonly using gradient-based optimization techniques. Owing to this learning mechanism, ANNs are well suited for capturing nonlinear relationships and have been widely applied in areas such as forecasting, pattern recognition, image analysis, and signal processing. Their performance improves with larger datasets and sufficient training, though they may require significant computational power and careful parameter tuning. Overall, ANNs are a powerful and flexible tool in machine learning, enabling intelligent decision-making and predictive modeling across various domains.

$$y\hat{} = f^{(L)}(\mathrm{w}^{(L)} f^{(L-1)}(\cdots f^{(1)}(\mathrm{w}^{(1)} x + b^{(1)})) + b^{(L)}) \quad (2)$$

Here, $f^{(l)}(\cdot)$ denotes the activation function, while $\mathrm{w}^{(l)}$ and represent the weight matrix and $b^{(l)}$ represent bias vector for layer l.

An ANN with a MLP architecture is a deep learning model designed to capture complex, non-linear relationships within data. Equation (2) outlines the adopted MLP configuration, which employs an input layer followed by two hidden layers of 64 and 32 neurons and a concluding output layer. Each neuron processes input signals by multiplying them with assigned weights, applying an activation function like ReLU, sigmoid, or tanh, and transmitting the resulting output to the subsequent layer. The network is trained using backpropagation combined with optimization algorithms like gradient descent to minimize the prediction error. In this work, the ANN(MLP) model achieved exceptional performance with an R^2 score of 1.0000, RMSE of 16 MW, and MAE of 10 MW in regression analysis, while its classification accuracy reached 98.5% across three demand levels (Low, Medium, and High). These results describe the model ability to approximate complex functions and deliver precise short-term power generation forecasts. Despite requiring higher computational effort and parameter tuning, the ANN(MLP) remains a powerful and reliable approach for intelligent energy forecasting applications.

The companion regression plot in Fig. 8 places every predicted power value virtually on the 45-degree reference line, revealing that the ANN reproduces actual grid loads with remarkable numerical fidelity across the full operating range.

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The ANN confusion matrix in the Fig. 9 shows the model classifies demand almost flawlessly 179 of 181 high-demand cases, 179 of 181 low, and 183 of 184 medium land on the diagonal, leaving only a few stray errors.

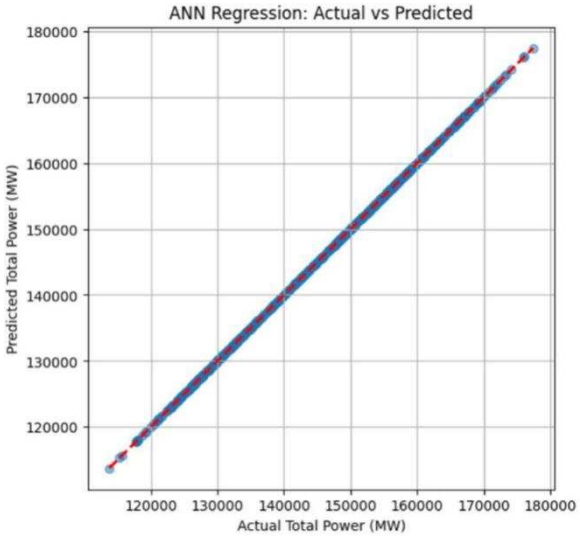


Fig. 8. Actual vs Predicted Power in Regression

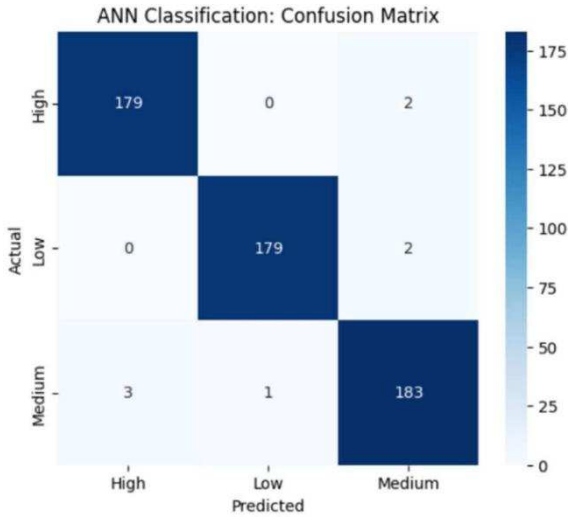


Fig. 9. Confusion Matrix

IV. RESULTS AND COMPARISON BETWEEN THE MODELS

The two models SVM and ANN were trained and optimized using the same dataset consisting of 2,742 daily Indian peak power generation records, ensuring a fair comparative evaluation. Hyperparameter tuning was conducted for both approaches: in the SVM, parameters such as the kernel type, penalty factor (C), and gamma were optimized, while in the ANN, the number of hidden layers, neurons per layer, learning rate, and training epochs were adjusted for best performance.

A. Performance of SVM:

The SVM with RBF kernel achieved high predictive accuracy after optimization. The regression model attained an R^2 of 0.9976, RMSE of 731 MW, and MAE of 452 MW, indicating excellent capability in modeling nonlinear relationships between input and output variables. For classification, the SVM achieved an impressive accuracy of 99.45%, effectively distinguishing between low, medium, and high demand categories. The close alignment of predicted and actual values confirms the robustness of the SVM in short-term power prediction tasks.

B. Performance of ANN:

The ANN employing a MLP architecture demonstrated even higher precision. Utilizing a dual hidden-layer structure with 64 neurons in the first layer and 32 in the second., the optimized ANN model achieved a perfect fit (R^2 = 1.0000), with exceptionally low errors RMSE of 16 MW and MAE of 10 MW. The classification model achieved 98.5% accuracy across all demand levels. The regression results exhibited nearly complete overlap between predicted and actual power values, validating the MLP's superior learning capability for complex nonlinear dependencies within the dataset.

C. Comparison of Models:

The comparative results shown in Fig. 10 confirm that the ANN(MLP) model significantly outperforms the SVM(RBF) in both regression and classification tasks. Although the SVM shows marginally higher classification accuracy, the ANN achieves much lower error values and superior generalization, making it more suitable for accurate short-term power forecasting. The SVM remains efficient and computationally less demanding, which is advantageous for smaller datasets or rapid deployment. However, for high-precision forecasting in renewable-integrated smart grids, the ANN model is the recommended approach due to its enhanced stability, adaptability, and predictive accuracy. The SVM(RBF) model compiled in 7 seconds, demonstrating faster execution, while the ANN(MLP) model required 47 seconds, indicating higher computational complexity but delivering superior accuracy and predictive performance.

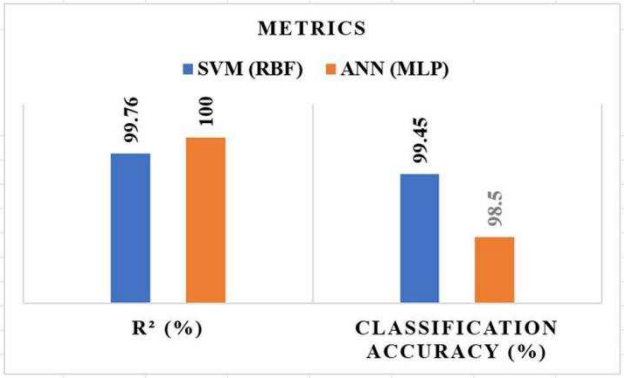


Fig. 10. Performance Comparison

V. CONCLUSION

This study conducted a comprehensive comparison between SVM based on a RBF and ANN based on a MLP architecture for short-term power generation forecasting in renewable-integrated grids. The optimized SVM with an RBF kernel achieved strong predictive performance (R^2 = 0.9976, RMSE = 731 MW, MAE = 452 MW, and accuracy = 99.45%), indicating effective classification but moderate regression precision. In contrast, the MLP-based ANN achieved a perfect correlation (R^2 = 1.0000) and

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substantially lower errors (RMSE = 16 MW, MAE = 10 MW) with 98.5% classification accuracy, demonstrating its superior capability to model nonlinear and dynamic patterns in power data.Overall, the findings confirm that while SVM offers computational efficiency and is suitable for smaller or less complex datasets, the ANN–MLP model provides exceptional accuracy, stability, and scalability making it the preferred choice for real-time grid forecasting and renewable integration. Future work should explore hybrid and explainable AI models, transformer-based deep learning architectures, and data-driven optimization frameworks to enhance transparency, adaptability, and robustness in smart grid applications, ultimately contributing to more reliable and sustainable energy management systems.

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